



TECHNICAL MAINTENANCE MANUAL

Water Pump Maintenance Manual - UH-PUMP-403

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Client	UrbanHive Facility Technologies Pvt. Ltd.
Site	GreenPark Meridian Residency, Whitefield, Bengaluru
Effective date	2024-07-01
Review cycle	Annual or after any major incident

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1. Equipment Overview and Technical Specifications

End-suction centrifugal booster pump for domestic water header pressure maintenance at GreenPark Meridian. Stainless impeller, mechanical seal, pressure transducer, and VFD control maintaining 2.8-3.2 bar setpoint.

Model code: UH-PUMP-403

Asset registry: Record serial number and installation tower in AMIS before first energization.

Electrical supply: 415V 3-phase 7.5 kW, IP55 motor, VFD with pressure PID loop

Environmental limits: Ambient 0-45°C operational; store spare parts below 35°C and 60% RH.

Standards: UrbanHive engineering bulletin UH-ENG-AMIS-2024; applicable Indian Standards as cited per asset class.

Documentation package: This manual, wiring diagram sheet WD-UH-PUMP-403, commissioning checklist CC-028, and spare parts catalogue SP-UH-PUMP-403.

Technicians must verify nameplate data matches AMIS asset record UH-PUMP-403 before applying procedures in this document. Discrepancies must be logged as a data quality ticket to UrbanHive Data Engineering within one business day.

2. Installation Guidelines

Install on inertia base with flex connectors on suction and discharge. Prime pump before first start. Calibrate pressure transducer 0-6 bar against dead-weight tester. Register curve in AMIS.

Pre-commissioning checks:

- Verify incoming power quality: voltage balance within 2%, THD < 5% on dedicated feeder.
- Confirm earthing resistance <= 1 ohm at equipment location; bond to society main earth bar.
- Inspect shipping damage; photograph defects for vendor warranty within 48 hours.
- Complete mechanical alignment, torque logs, and insulation resistance tests per UrbanHive QC form UH-QC-INST-02.

Post-installation: Upload signed commissioning certificate, as-built photos, and warranty start date to AMIS. Tag physical asset with QR code linking to this manual revision v1.0.0.

3. Preventive Maintenance Schedule

UrbanHive mandates the following intervals for GreenPark Meridian. Adjust only with written approval from Head of Maintenance.

- Daily: suction strainer ?P log, abnormal noise check.
- Weekly: gland leakage inspection, motor bearing temperature IR scan.
- Monthly: coupling alignment check, seal flush flow verification.
- Quarterly: vibration analysis; impeller wear visual via drain port.
- Annual: mechanical seal replacement; bearing grease per SKF chart.

Documentation: Each PM visit must generate a service log in AMIS with parts consumed, meter readings, and pass/fail checklist. Missed PM windows trigger automatic escalation to Facility Manager after 72 hours.

4. Troubleshooting Guide

Use isolation procedure: observe symptom ? confirm with meter/test ? apply remediation ? functional test ? document in AMIS.

If fault persists after two visits, escalate to UrbanHive vendor support with fault code and photos.

Fault 01 - Code P101

Symptom: Failure to build pressure

Remediation: Verify suction valve open; prime casing; check for air lock in rising main.

Fault 02 - Code P203

Symptom: Cavitation noise at impeller

Remediation: Increase suction head; clean strainer; reduce flow below NPSH available.

Fault 03 - Code P310

Symptom: Mechanical seal leak > drip rate

Remediation: Adjust seal flush; replace seal faces if scoring visible.

Fault 04 - Code P405

Symptom: VFD overcurrent trip

Remediation: Measure motor current imbalance; inspect impeller for obstruction.

Fault 05 - Code P512

Symptom: Pressure hunting ± 0.4 bar

Remediation: Retune PID gains; verify transducer snubber not clogged.

Fault 06 - Code P601

Symptom: Motor bearing over-temperature

Remediation: Grease per interval; check alignment; replace bearing if play > 0.05mm.

Fault 07 - Code P708

Symptom: Dry run protection false trip

Remediation: Clean electrodes in sump; verify cable shield grounding.

Fault 08 - Code P802

Symptom: Coupling rubber spider cracked

Remediation: Replace spider element; realign within 0.1mm/100mm.

Fault 09 - Code P901

Symptom: Check valve chatter

Remediation: Inspect disc seat for debris; verify spring tension spec.

Fault 10 - Code P1004

Symptom: Unusual vibration at 2 $\times$ line frequency

Remediation: Tighten base bolts; inspect foundation grout cracks.

Fault 11 - Code P1102

Symptom: Header tank overflow linked event

Remediation: Test float switch continuity; verify stop setpoint in PLC.

Fault 12 - Code P1203

Symptom: Discharge pressure below SLA 2.5 bar

Remediation: Parallel standby pump test; inspect non-return valve stuck closed.

5. Spare Parts List

Maintain minimum stock at GreenPark Meridian central store (Basement B1, Room ST-04).

Part No.	Description	Min stock
SP-P01	Mechanical seal kit	4
SP-P02	Coupling spider	6
SP-P03	Pressure transducer 0-6 bar	2
SP-P04	Strainer basket SS316	3
SP-P05	VFD cooling fan	2
SP-P06	Gland packing set	5

Reorder when stock falls below minimum. Use UrbanHive procurement portal PR-UH-OPS for approved vendors only.

6. Safety Warnings and Compliance

Isolate electrical and lock-out before seal work. Drain line pressure before disassembly. Confined space rules apply for basement pump rooms. UrbanHive permit required for welding on discharge header.

Permit requirements: Hot work, confined space, work-at-height, and electrical isolation require digital permit in AMIS before start.

PPE minimum: Safety shoes IS 15298, helmet IS 2925, eye protection IS 5983, gloves appropriate to task.

Incident reporting: Any injury, near-miss, or property damage - complete IR-01 within 30 minutes; notify security desk ext. 9911.

7. Manufacturer Contact and UrbanHive Support

OEM technical support: +91-80-5555-0028 (08:00-20:00 IST)

Email: support-uh-pump-403@urbanhive-vendor.example.com

UrbanHive Ops escalation: maintenance@urbanhiveops.example.com - Priority 1 response within 4 business hours

AMIS helpdesk: Log ticket category `Water Pump` with asset_id from tower registry

Emergency (life safety assets): 24x7 hotline +91-80-5555-0911

Revision history: v1.0.0 - Initial release for GreenPark Meridian AMIS integration and UrbanHive contractor certification programme.

End of document MAN-028